

Dynamic and Static Workout of In Vitro Skeletal Muscle Tissue through a Weight Training Device

Byeongwook Jo, Kentaro Motoi, Yuya Morimoto, and Shoji Takeuchi*

Enhancing muscle strength through workouts is a key factor in improving physical activity and maintaining metabolic profiles. The controversial results concerning the impacts of weight training often arise from clinical experiments that require controlled experimental conditions. In this study, a weight training system for a muscle development model is presented, which is capable of performing weight training motions with adjustable weight loads. Through the implementation of cultured skeletal muscle tissue with floating structures and a flexible ribbon, both isotonic (dynamic change in muscle length) and isometric (static in muscle length) exercises can be performed without the deflection of the tissue. Quantitative analysis of contraction force, changes in metabolic processes, and muscle morphology under different weight training conditions demonstrates the effectiveness of the proposed system. Our proposed system holds potential for establishing effective muscle development and for further applications in rehabilitation training methods.

1. Introduction

Accounting for 40% of the human body mass, skeletal muscles are fundamental to not just physical movement but also to

maintaining metabolic profiles.^[1] Enhancing muscle development through regular exercise is a critical concern, relevant both in academic research and practical daily activities. Applying resistance to the muscle through weight training is one of the most practical methods to promote muscle development due to its ability to provide progressive overload, targeted training, and widespread accessibility.^[2,3] Studies in sport and medical sciences provide optimal weight training conditions for effective muscle development,^[4] but clinical experiment methods often yield controversial conclusions due to variations in experimental conditions including age, sex, races, and athletic skill level.^[5–8] Additionally, directly evaluating the specific responses of muscle development is challenging due to potential

influences from other organs such as skin, cartilage, and bone. Recent advances in tissue engineering offer solutions to these challenges by fabricating physiologically relevant skeletal muscle tissues.^[9,10] The control over tractional force and the ability to quantify drug responses present a wide range of applications from biohybrid robots^[11–13] to micro-physiological systems.^[14–16] Electrical pulse^[17–19] and mechanical stretch^[20,21] based training approaches, designed to mimic muscle training for hypertrophy and metabolic flux maturation, suggest potentials in personalized treatments for rehabilitation or training programs. However, previous in vitro muscle models feature only isometric training (static in muscle length) with fixed substrates holding the muscle. Since most of the dynamic motions require isotonic exercise (change in muscle length), for further comprehensive analysis on the effect of weight training, in vitro muscle models should be capable of applying various exercise conditions under active contraction.

In this paper, we propose a weight training system for a muscle development model with implementation of cultured skeletal muscle tissue capable of achieving the weight training motions (**Figure 1**). Through the floating structures and adjustable weights connected through the flexible ribbon, weight training motion can be applied to the muscle under electrical stimulation without the deflection of the muscle tissue. By adjusting weight load, both isotonic and isometric exercises are demonstrated with the comparison of an increase in muscle contraction force, alignment, and muscle fiber diameter. Also, differences in glucose consumption and lactate production in muscle tissue between the training and non-training group over a 15 d culture period show potentials in clarifying the effects of weight training on cellular metabolism. By offering highly controllable weight training

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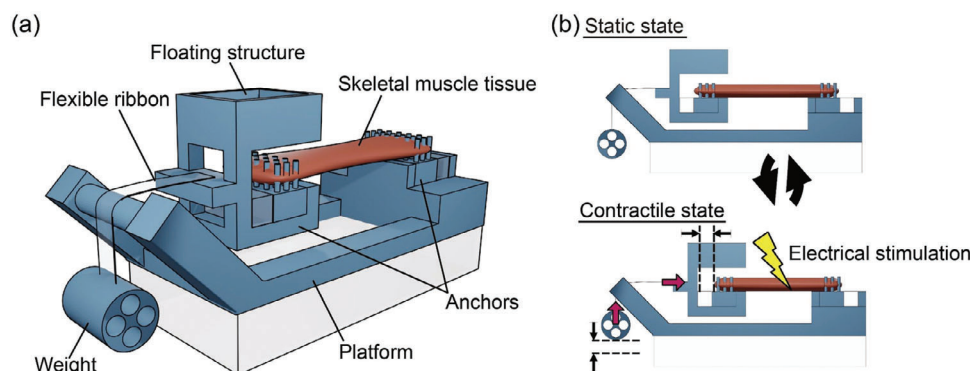


Figure 1. a) Conceptual illustration of the weight training device with a fabricated skeletal muscle tissue. b) Schematic illustration of the weight training through the contraction of the fabricated skeletal tissue by applying electrical stimulation.

conditions and addressing clinical challenges, our system sets the stage for more precise and standardized approaches to muscle development research.

2. Results

2.1. Characteristics of the Weight Training System

The weight training movement proposed in this study requires the contraction of skeletal muscle tissue accompanying vertical movement of weights. By seeding the cell-suspended hydrogel structure around the anchors followed with 10 d of culture after the differentiation, skeletal muscle tissue, which could be manipulated with tweezers, was produced (Figure 2a and Figure S1, Supporting Information). Under electrical stimulation with different pulse frequency, the fabricated skeletal muscle tissue showed twitch at 1 Hz, and tetanic contraction at frequencies over 5 Hz (Figure 2b), showing similar characteristics to the skeletal muscle tissues described in previous works.^[11,12] Through the contractile force measurement using a PDMS cantilever (Figure S2, Supporting Information), the maximum contractile force of 3.6×10^{-1} and 1.3 mN was obtained under twitch and tetanic contraction, respectively (Figure 2c). Also, electrical fields over 1.5 V mm^{-1} induced convergence of the contractile force at both twitch and tetanic contractions. These results indicate the capability of controlling the muscle contractions under desired strength and frequency which is critical to apply weight training conditions. The implementation of the fabricated skeletal muscle tissue to the weight training system with one fixed end and one free end connected to the weight applied constant tension to the muscle and managed a state of equilibrium (Figure 2d and Figure S3, Supporting Information). Under the electrical stimulation on the skeletal muscle tissue, the lift-up motion of the weight training was observed (Figure 2e and Movie S1, Supporting Information). Also, owing to the floating structure which applied constant buoyancy to the skeletal muscle tissue, the balance of the tissue was maintained during the lift-up motion. The skeletal muscle lifted the weight by 0.43 mm, compared to only 0.21 mm in samples without the floating structure (Figure 2f). Without the floating structure, the contractile force induces the displacement along the gravitational direction which results in losses in lift-up distance of the weight (Figure 2g,h). This weight training motion, derived from the active contraction of skeletal muscle

tissue, clearly differs from conventional training systems with fixed endpoints^[15,22,23] or passive mechanical stretch,^[21,24–26] making it more suitable for mimicking physiologically relevant conditions of the skeletal muscle tissue during training exercises.

2.2. Evaluation of Lifting Distance under Different Weight

Through the weight training on skeletal muscle tissue for 5 d after 11th day of the tissue fabrication, the effects of weight training were achieved by evaluating the lifting distance on different weights. The applied electrical pulses (electrical field, 3.0 V mm^{-1} ; frequency, 50 Hz; pulse width, 10 ms), with repetition of 1 s on and 4 s off, for a duration of 18 min each day with different weights enabled the simulation of daily weight-training motions. The lifting of the weight on day 5 of training is shown in Figure 3a,b when using 100 mg and 200 mg weights, which corresponds to 75% and 150% of maximum contractile force generated from the muscle. 100 mg of weight was clearly lifted in response to the electrical stimulation, while changing the muscle length at the same time (Movie S2, Supporting Information). On the other hand, significant lifting motion was not observed with a 200 mg weight, since the gravitational force applied to the weight was around 2.0 mN, which exceeded the contractile force of the skeletal muscle tissue (Figure 3c). This constant length of the skeletal muscle tissue under the contraction shows the isometric training motion. These results demonstrate that changing the weight can lead to different weight training movements such as isotonic and isometric training. When the lifting distance of the weight on samples with and without the 5 d training was compared, a significant difference in lifting distance was observed, indicating an increase in contractile force owing to the weight training (Figure 3d). The moving distance of the weight for each weight (0, 100, 200, and 300 mg; 0, 75, 150, 224% of maximum contractile force, respectively) were changed according to the training days (Figure 3e). Additionally, Figure 3f shows changes in moving distance of the weight on the first and fifth days of training. Weight 0 and 100 mg tended to increase the lifting distance as the training proceeds, with a significant increase of 370% and 324% observed respectively. Conversely, the weights of 200 and 300 mg did not exhibit any significant trend towards increased distance with increase in training days. Specifically, the

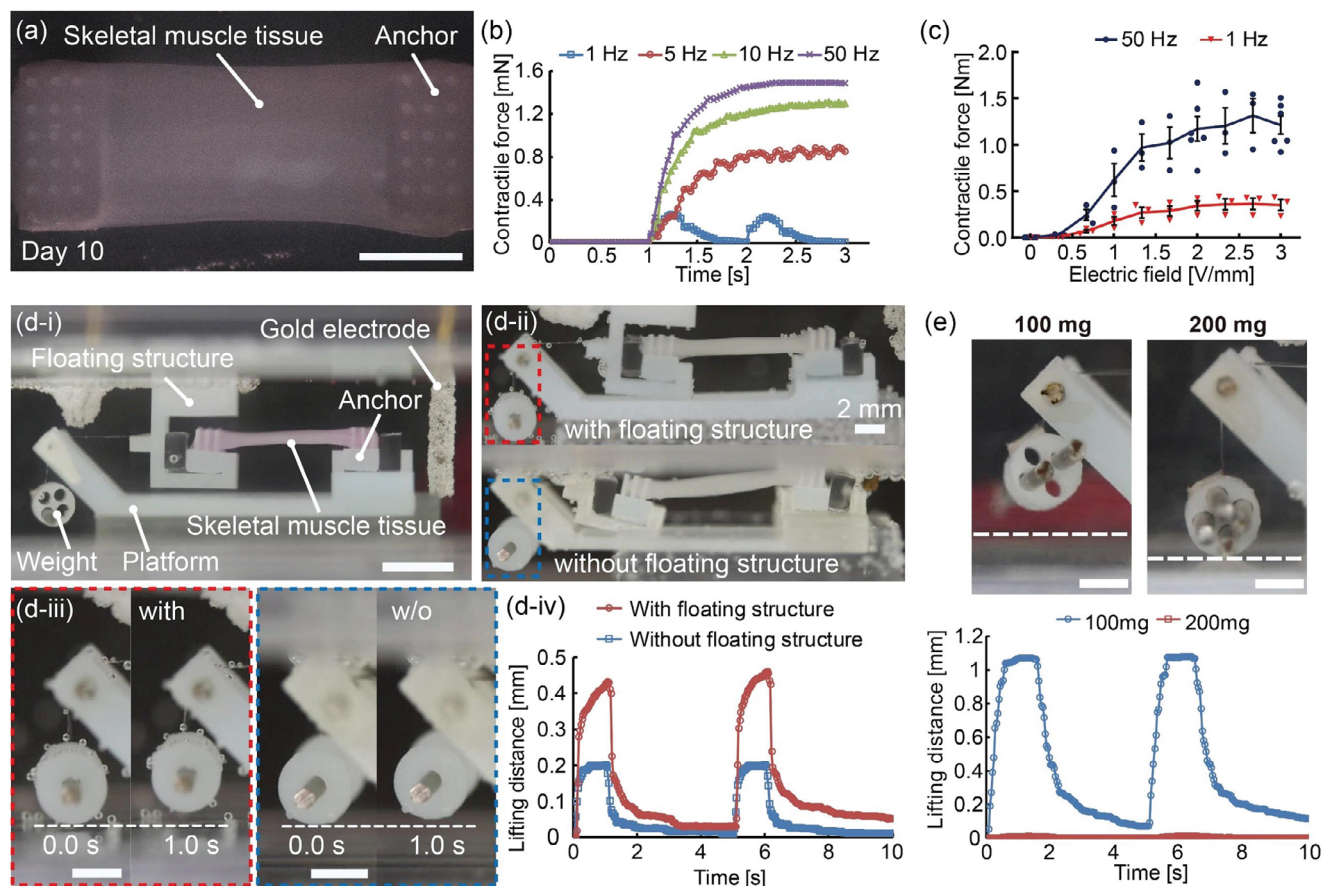


Figure 2. a) Microscopic picture of the fabricated skeletal muscle tissue. 3D printed anchors enable handling of the tissue which can be mounted to the weight training system. Scale bar, 3 mm. b) Contraction of the skeletal muscle tissue under electrical stimulation with different frequency. Applied electrical field was 3.0 V mm^{-1} with 2 ms pulse width. c) Contraction of the skeletal muscle tissue under different electrical field. ($n \geq 3$, mean $\pm$ SEM). d) Photograph of the weight training system mounted with the skeletal muscle tissue. e) Photograph of the weight training system mounted with the skeletal muscle tissue. Scale bar, 5 mm.

300 mg weight moved 0.11 mm or less, which is less than 1% of the length of the skeletal muscle tissue indicating that isometric training was applied throughout the 5 d training period. The total lifting distance of the weight during the training period was varied for each weight (Figure 3g). The results show that the total moving distance of the weight reached its maximum around 50 to 100 mg and tended to decrease as the weight increased. During the training, tissue length in static state was evaluated and compared with the length of the tissue during the culture (Figure 3h). In case of the samples without weight (0 mg), length of the skeletal muscle tissue shrank due to the tractional force, whereas weight over 100 mg kept the length. Since the contractile force of the skeletal muscle tissue significantly decreases after the shrinkage in length,^[11] the weight selection during the training should be considered to exceed the tractional force of the tissue. Regardless of different training conditions, cell density of the skeletal muscle tissues were in $1.0 \times 10^8 \text{ cells cm}^{-3}$ order showing comparable values to previous study.^[27] Since cell enucleation or necrosis results in low cell density of the tissue, this result indicates that the weight training did not affect the cell viability (Figure S4, Supporting Information).

2.3. Effects of Weight Training on Contractile Force and Change in Metabolic Process of Skeletal Muscle Tissues

To evaluate the effect of weight training, the contractile force, glucose consumption and lactate production of the skeletal muscle tissue were evaluated. Through the PDMS cantilever, the contractile force of the skeletal muscle tissue was measured under different weights. Compared to the contractile force of $1.2 \pm 0.2 \text{ mN}$ with untrained skeletal muscle tissue, weight trained samples showed higher contractile force maximum up to $1.8 \pm 0.3 \text{ mN}$ at 100 mg (Figure 4a). Also, we compared the weight trained skeletal muscle tissues at 100 and 300 mg, which resembles isotonic and isometric exercise as mentioned above. The contractile force for 100 mg (isotonic) and 300 mg (isometric) showed 151% and 117% compared to untrained samples, respectively (Figure 4b). The results from isometric exercise corresponded to the tendency reported in previous studies.^[24,28] To be noted, it is the first time to compare the isotonic and isometric exercise with cultured skeletal muscle tissue which provides clues to distinguish the effects of different exercise types that showed controversial results in clinical experiments.^[5–8]

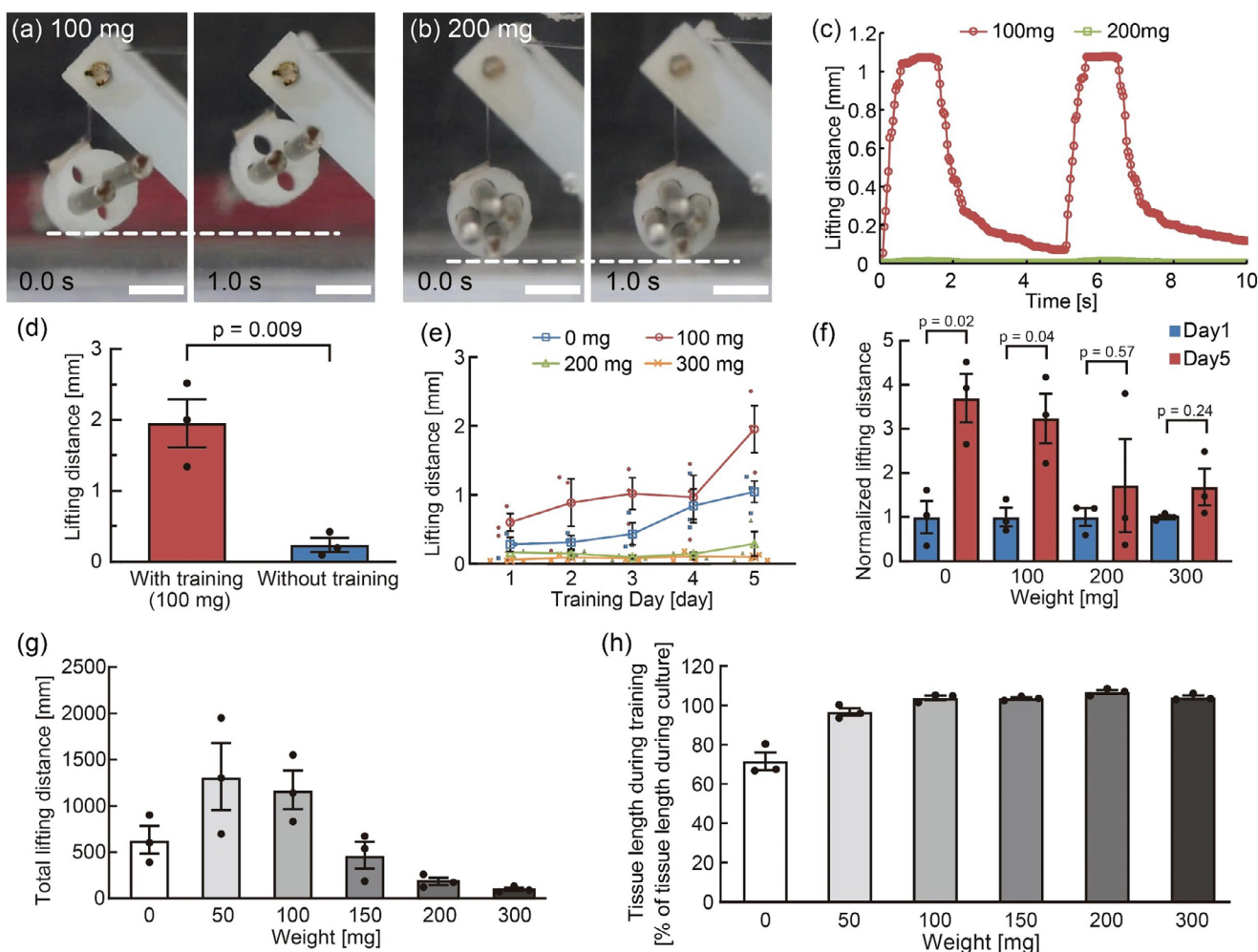


Figure 3. a,b) Sequential images of the lifting of the weight through the contraction of the skeletal muscle tissue when using 100 mg and 200 mg weights. Scale bars, 2 mm. c) Lifting distance of the weight through the contraction of the skeletal muscle tissue by applying electrical stimulation. d) Lifting distance of the weight on skeletal muscle tissues with and without the weight training. ($n = 3$). e) Changes in lifting distance of the weight for each weight along training days. ($n = 3$). f) Normalized lifting distance of the weight on the first and fifth days of training. ($n = 3$). Values were normalized to the first day of the training. g) Changes in the total lifting distance of the weight during the training period at each weight. Each training period was 18 min. ($n = 3$). h) Length of the tissue in static state during the training. Values were shown in percentage of tissue length during the culture. ($n = 3$). Data are presented as mean $\pm$ SEM. Statistical significance was determined by t -test assuming equal variance.

To evaluate the effects of weight training on the change in metabolic process of the skeletal muscle tissue, glucose consumption and lactate production within the culture media was measured after 24 h from day 5 of the weight training. Compared with the untrained samples, the weight-trained sample showed higher glucose consumption and lactate production of 125% and 163%, respectively (Figure 4c,d). From these results, it was confirmed that the weight training not only significantly increased the contractile force but also changed the metabolic process of the skeletal muscle tissue.

2.4. Effects of Weight Training on Morphology of Skeletal Muscle Tissue

The effect of weight training on the orientation of skeletal muscle tissue was evaluated by analyzing confocal microscopic images

of three types of samples: without training (control), weight training at 100 mg (isotonic), and 300 mg (isometric) (Figure 5a–c). Staining the cytoskeleton of a sample with phalloidin enables evaluating the orientation of skeletal muscle tissue from the long-axis confocal microscopy images. The directional distribution of skeletal muscle fibers shows that skeletal muscle fibers were oriented in the direction of 90° , i.e., the long axis of skeletal muscle tissue, in all samples (Figure 5d–f). The orientation of each sample indicates that there was a significant difference between the control sample and both weight training samples (isotonic and isometric). However, there was no significant difference between the isotonic and isometric training groups (Figure 5g). These results suggest that weight training using the proposed method is effective in improving the orientation of skeletal muscle fibers, which may lead to an improvement in muscle maturity. In the confocal images of the skeletal muscle tissues, a stripe-like distribution of α -actinin (Figure 5h) and myosin heavy chain

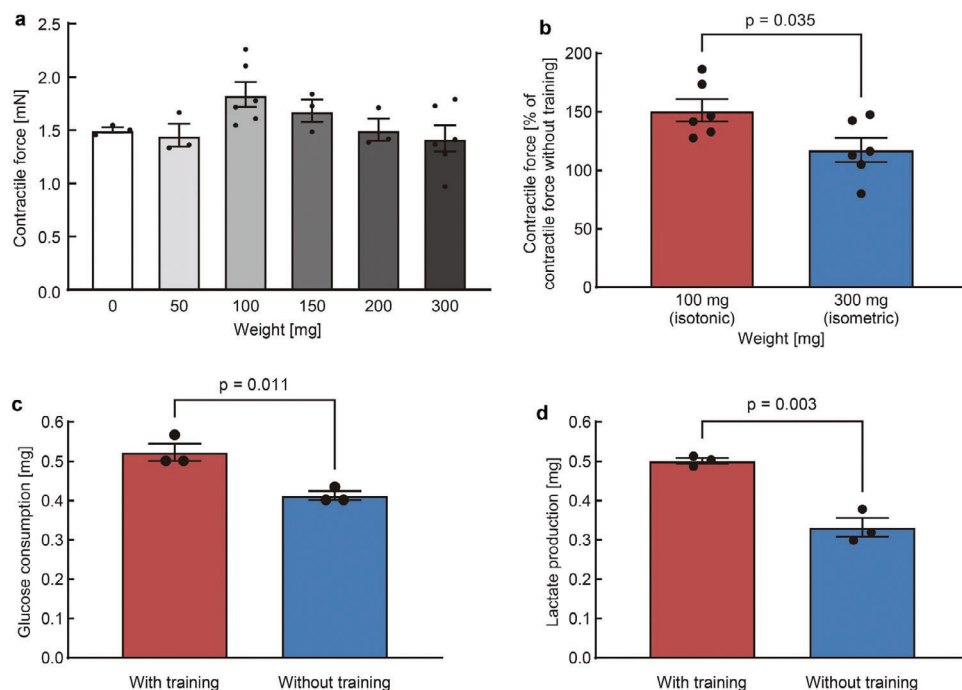


Figure 4. a) Changes in contractile force of the skeletal muscle tissues after weight training with different weights. ($n \geq 3$). b) Contractile force of skeletal muscle tissues with weight training at 100 mg and 300 mg. Contractile force was normalized by the contractile force of the skeletal muscle tissue without training. ($n = 6$). c) Changes in glucose consumption between the skeletal muscle tissue with and without training. ($n = 3$). d) Changes in lactate production between the skeletal muscle tissue with and without training. ($n = 3$). Data are presented as mean $\pm$ SEM. Statistical significance was determined by t -test assuming equal variance.

(Figure S5, Supporting Information) was observed, indicating the morphological characteristics of the myofibers that are found in native skeletal muscle tissue. The length of the sarcomeres was 2.4 ± 0.2 , 2.3 ± 0.3 , and 2.6 ± 0.3 μm for the control, 100 mg (isotonic), and 300 mg (isometric), respectively, showing similar values to native skeletal muscle tissue of 2.6 μm [29] (Figure 5i). However, there were no significant differences between the training and the non-training groups. Finally, the effect of weight training on myofiber diameter was evaluated. Myofiber diameters were measured from immune-stained microscopic images of frozen sections of each sample (Figure 6a–d). Muscle fiber diameters were 8.6 ± 2.1 , 10.3 ± 3.2 , and 9.3 ± 2.6 μm for the control, 100 mg (isotonic), and 300 mg (isometric) samples, respectively, which showed similar values as reported in previous studies. [17,30,31] Myofiber diameters in the 100 mg and 300 mg samples increased by 20% and 8%, respectively, compared to the control sample. These results indicate that weight training significantly increased the diameter of myofibers compared to the non-training samples (Figure 6e). The average myofiber diameters for isotonic training was 11% higher than that of isometric training, however, further analysis and investigation with large numbers of samples might be needed for validation.

3. Discussion

This study demonstrates a weight training motion of cultured skeletal muscle tissue through adjustable weight load with changes in muscle length under electrical stimulation. Functional and morphological analysis of a 5 d training group shows

a 29% higher contractile force and 11% thicker muscle fiber diameter in isotonic training compared to isometric training. Also, evaluations of glucose consumption and lactate production show the effects of weight training compared to non-training groups. These results indicate the extrinsic evaluation of the weight training effect, overcoming the limitations of clinical evaluation methods that show controversial results due to age, sex, race, and differences in training conditions.

Although training of the in vitro skeletal muscle tissues through electrical stimulation [19,24,32] or mechanical stretch [21,24–26] has been presented, previous works were limited to either fixed endpoints (isometric training) [15,22,23] or passive changes in muscle length. [21,24–26] Since most weight training motions, such as lifting and raising, require active contraction of the muscle along with changes in length, [33,34] performing an isotonic training is inevitable to achieve similar training conditions that occur in the human body. Also, passive mechanical stretching through external force might accelerate the maturation of myotubes, it does not mimic the muscle contraction metabolism of glucose consumption and lactate production. Owing to the floating structure and a movable endpoint, our proposed system can realize weight training motions with active changes in muscle length, also satisfying the muscle contraction metabolism.

There are several areas for improvement in the present system to further facilitate weight training mechanisms similar to weight training in the human body. The isotonic training demonstrated in this study can be specified into eccentric and concentric training which shortens and lengthens the muscle length, respectively, during the contraction. Both eccentric and concentric

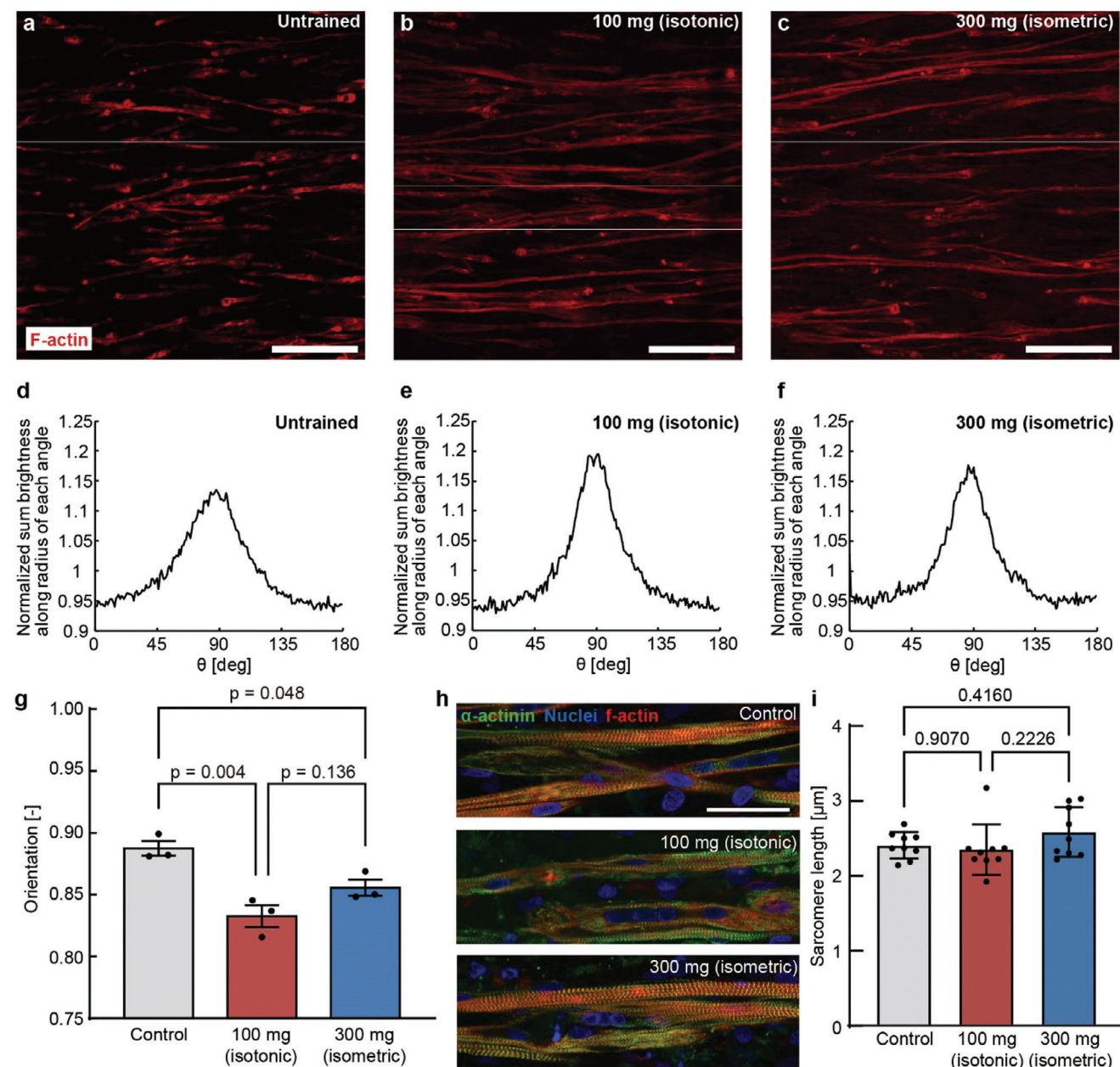


Figure 5. a–c) Long-axis confocal images of skeletal muscle tissues stained for filamentous actin (F-actin, red) without training or with weight training at 100 or 300 mg. Scale bars, 200 μm . d–f) Directional distribution of brightness of skeletal muscle tissues without training (Ctrl) or with weight training at 100 mg or 300 mg calculated by fast Fourier transform (FFT). g) Orientation of skeletal muscle tissues without training and with weight training at 100 and 300 mg. ($n = 3$). h) Long-axis confocal images of the skeletal muscle tissue stained for α -actinin (green) and cell nuclei (blue). Scale bar, 50 μm . i) Sarcomere length of the skeletal muscle tissue without training (Ctrl) or with weight training at 100 mg and 300 mg ($n = 9$). Data are presented as mean $\pm$ SEM. Statistical significance was determined by one-way ANOVA with multiple comparisons via Tukey's test.

training are included in most weight training, and recent clinical studies imply that each training results in different benefits in terms of muscle strength and injury risks.^[35,36] Therefore, it is crucial not only to mimic training motions in the human body but also to quantitatively analyze the difference in training types. The present study focuses on concentric training to highlight the functional aspects of the proposed system, but since the fabricated skeletal muscle tissue responds to the electrical stimulation with different frequencies and amplitudes, eccentric training can

also be applied to our system with additional adjustments. Also, it is worth comparing gene expression rates of myosin heavy chain (MYH) isoforms to investigate different fiber types, such as MYH1 (fast-twitch), and MYH7 (slow-twitch). Since human primary cells (fetal) were used as a proof of concept in the current weight training system, MYH3 (embryonic) and MYH8 (neonatal) expression rates are likely to be high.^[37,38] For future applications, using human induced pluripotent stem cells (iPSCs) hold more potential in terms of scalability, genetic consistency,

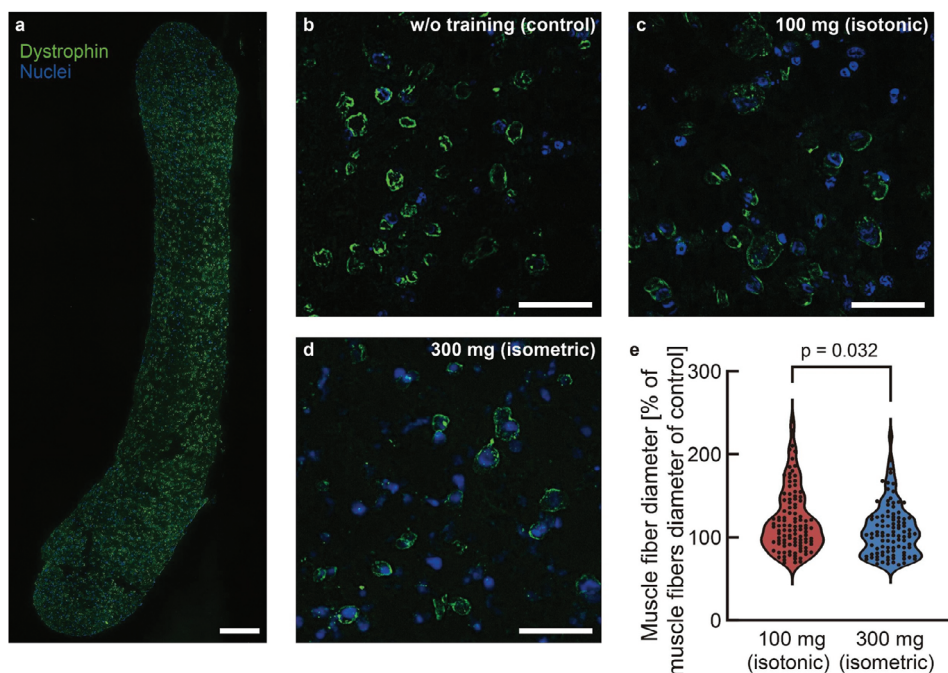


Figure 6. a) Short-axis sectional image of the skeletal muscle tissue stained for dystrophin (green) and cell nuclei (blue). Scale bar, 200 μm . b–d) Zoomed-in short-axis sectional images of skeletal muscle tissues without training (Ctrl) or with weight training at 100 or 300 mg. Scale bars, 20 μm . e) Muscle fibers diameter of skeletal muscle tissue with weight training at 100 and 300 mg. Muscle fibers diameter was normalized by the muscle fibers diameter of the skeletal muscle tissues without training. ($n \geq 90$ from 3 individual samples). Data are presented as mean $\pm$ SEM. Statistical significance was determined by a two-sided Mann-Whitney test.

and disease modeling. Next, within the present study, contractile force of the skeletal muscle tissue was induced through electrical stimulation, but an alternative approach involving nerve stimulation for muscle contraction could be considered. To mimic the contraction mechanism that occurs in the human body, a series of action from neuromuscular junction to sarcoplasmic reticulum which releases calcium ion channels is indispensable. Implementing the co-culturing method of skeletal muscle tissue with motor neurons [14,39,40] to the present study could establish an environment more akin to in vivo conditions. Also, recent advances in opto-genetics allow induction of muscle contraction through optical stimulation [44–43] which has a lower risk of contamination, electrolysis of culture media, and capability of remote control. In addition, within the present system, skeletal muscle tissue is directly connected to rigid 3D printed structures, but skeletal muscles in vivo are linked to bones via tendons preventing muscle damage by absorbing impact and facilitating efficient force transmission, which significantly influence the movement performance. To achieve a high degree of biomimetic training, it would be necessary to construct tendons at both ends of skeletal muscle tissue. Since recent commercial 3D printers offer flexible resins with various degrees of stiffness, applying these techniques to the current system would provide practical developments.

4. Conclusion

This research proposed a weight training system capable of simulating length changes in cultured skeletal muscle tissue when

contractile force is applied through electrical stimulation. By utilizing a floating structure and adjustable weight, implementation of changes in muscle length without deflection was achieved which enabled demonstrating isotonic training for the first time in cultured skeletal muscle tissue. The comparison of isometric and isotonic in terms of contractile strength and morphological differences is expected to make a contribution to the field of sports science with more stable and quantified results which have been relying on clinical experiments. Moreover, our approach holds promise for personalized medicine applications. By using cultured skeletal muscle tissue, adjusting exercise regimens to individual cellular responses is possible, paving the way for customized rehabilitation and training programs. Especially in clinical settings, where patient specific muscle conditions, ranging from atrophy to specific athletic training requirements, could be addressed more effectively. In addition, from the glycolysis and lactate production analysis between training and non-training groups, we believe that this weight training system holds high applicability not only in biological evaluation but also in drug testing, such as quantitative evaluation of muscle strength enhancement by drugs.

5. Experimental Section

Cell Culture: Human skeletal muscle cells (150K-05f, Cell Applications) with passages under 5 were used throughout the experiment. SkGM-2 bullet kit (CC-3245, Lonza) was used as growth media and the media was changed every other day. For differentiation media, DMEM low glucose supplemented with 2% (v/v) horse serum, and 1% (v/v) penicillin streptomycin was used.

Culture Device Preparation: The culture device used for fabricating and culturing skeletal muscle tissue consists of a culture table and anchors. The culture table shapes the skeletal muscle tissue, while the anchors secure the tissue at both ends during the culture. To prepare the culture table, conventional soft-lithography method was used. First, a master mold was prepared through a 3D printer (Agilista-300, Keyence) and the surface of the 3D printed parts was covered with 2 μm of parylene (Parylene-C, Specialty Coating Systems). Next, polydimethylsiloxane (PDMS, Dow) with the mixture of the base and crosslinker with a 10:1 ratio was poured into the master mold and heated at 75 $^{\circ}\text{C}$ for 90 min to form a culture table. In order to prevent cell adherence on the culture table, 0.5 wt% 2-methacryloyloxyethyl phosphorylcholine (MPC) polymer (Lipidure, NOF) diluted in ethanol was poured on the surface and heated at 65 $^{\circ}\text{C}$ for 90 min. The anchors were also 3D printed (DigitalWax 028J, DWS) and covered with 2 μm of parylene to improve biocompatibility. After sterilization through UV and ozone sterilizer (E330110, Gelantis) for 15 min, anchors were suspended in 50 $\mu\text{g mL}^{-1}$ of fibronectin (068-05703, Wako Pure Chemical) diluted in phosphate-buffered saline (PBS) overnight to improve adhesion of the skeletal muscle tissue. The anchors were then mounted to culture table to form a culture device.

Skeletal Muscle Tissue Fabrication Process: To fabricate skeletal muscle tissue, cell suspension solution and hydrogel solution were prepared. For cell suspension solution, cells were detached from culture dish with trypsin/EDTA (T4174, Lonza) and centrifuged at 200G for 5 min. After removing the supernatant, 68 μL of growth media and 8 μL of thrombin (T7009, Sigma-Aldrich) solution (50 units mL^{-1}) were added and carefully mixed by pipetting. The hydrogel solutions were prepared by mixing 44 μL of growth media, 40 μL of fibrinogen (F8630, Sigma-Aldrich) solution (20 mg mL^{-1}), and 40 μL of Matrigel (356231, Corning) or Cultrex BME (3445, R&D Systems), which were added and mixed. The whole preparation process was conducted on ice to prevent the hydrogels from solidifying. The cell suspension solution and hydrogel solution were then mixed with a pipette and immediately seeded on the culture device and incubated for 10 min to form a cell suspended hydrogel structure. The total cell concentration of the cell suspended hydrogel was 25 million cells mL^{-1} . Since the cell suspended hydrogel structure showed drastic compaction of the width due to tractional force of the cell, growth media was replaced with differentiation media to mature into skeletal muscle tissue after day 2 of culture (Figure S6, Supporting Information). To prevent dissolving of fibrin by serum, 1.5 and 2.0 mg mL^{-1} of 6-aminocaproic acid (A2504, Sigma Aldrich) were added to growth and differentiation media, respectively.

Weight Training System Fabrication and Assembly: The weight training device consists of a PDMS base, a platform, a floating structure, a weight and a flexible ribbon. The PDMS base was prepared to mount weight training device to the culture dish which was fabricated through the soft-lithography method as mentioned above. For the fabrication of the platform, floating structure and holder for weights, a 3D printer was used and covered with 2 μm of parylene. The platform works as a holder for the skeletal muscle tissue and a support for flexible ribbon during weight training. To reduce the friction while the flexible ribbon is moving, a glass tube (GD-1.5, Narishige) was inserted in the platform which was fixed with a stainless-steel rod (0.8 mm, Hikari). The floating structure was designed to apply buoyancy to the skeletal muscle tissue during the weight training for stabilizing the posture without the deflection. To apply buoyancy on the floating structure, an overhead projector (OHP) film was used to seal the cavity. The weight was used to apply load to the skeletal muscle tissue, which consisted of the weight holder and the stainless-steel rod. The flexible ribbon was prepared by cutting the 5 μm parylene sheet with a surgical knife which was formed on the glass plate coated with detergent for easy detachment. After UV and ozone sterilization, each part was assembled inside the clean bench to fabricate a weight training device and submerged in a culture media. The skeletal muscle tissue was then transferred to the weight training device and gold electrodes were placed for electrical stimulation to form a weight training system.

Weight Training Cycle: After 11 d of culture from the differentiation, the skeletal muscle tissue was trained with the proposed weight training system. The contraction of muscle was induced through an electrical stimulation system with 3.0 V mm^{-1} , pulse width 10 ms, 50 Hz for 1s followed

by 4 s of rest. The training was conducted for 18 min daily, for a total of 5 d. For visualization of the weight training system under the training, phenol red free culture media (040-30095, Wako Pure Chemical) was used.

Muscle Contractile Force Measurement: To measure the contractile force of the skeletal muscle tissue, the PDMS cantilever (Figure S2a, Supporting Information) was prepared. The skeletal muscle tissue was mounted to the PDMS cantilever (Figure S2b, Supporting Information) and the deflection of the cantilever during the contraction of the muscle was recorded using the microscope. Through the motion analyzer (VW-H2MA, Keyence), the deflection of the cantilever was analyzed and converted to contractile force of the skeletal muscle tissue with solid mechanic calculation (Figure S2c, Supporting Information).

Glucose Consumption and Lactic Acid Production Measurement: To evaluate the consumption of glucose and production of lactate of the skeletal muscle tissue, the biosensor (BF-7MX, Oujii Scientific Instruments) was used. The culture media before and after 24 h of culture were collected, and the concentration of target substances was measured using the biosensor according to the manufacturer's manuals. To remove remnant substances, culture dish was rinsed with PBS twice every time culture media was changed.

Cryosection: To analyze the morphological characteristics of the skeletal muscle tissue, cryosection was conducted. First, the tissue was fixed with 4% paraformaldehyde (PFA) (33111, Muto Pure Chemicals) for 30 min and rinsed with PBS for 3 times. For dehydration, the tissue was suspended in 10%, 20%, 30% (v/v) sucrose solution for overnight, consecutively. After removing the sucrose solution, the tissue was suspended with optical cutting temperature (OCT) compound for overnight, then liquid nitrogen was used to freeze. Finally, section of the tissue with 8 μm of thickness was obtained through cryostat (CryoStar NX50, Thermo Fisher Scientific).

Immunostaining: The skeletal muscle tissue was fixed with 4% PFA for 30 min and rinsed with PBS for three times. Next, the tissue was submerged in 0.2% (v/v) TritonX-100 (Alfa Aesar) solution for 5 min to permeabilize the cell membranes, followed by blocking with 5% (v/v) bovine serum albumin solution (BSA) for overnight. For sarcomere, myosin heavy chain, and dystrophin staining 0.2% (v/v) monoclonal anti- α -actinin antibody (Sigma-Aldrich), 0.1% (v/v) anti-myosin heavy chain antibody MF20 (Developmental Studies Hybridoma Bank), and 0.5% (v/v) anti-dystrophin antibody (Abcam) were used, respectively. After suspending the tissue overnight, the tissue was then incubated in 0.25% (v/v) Alexa Fluor conjugated anti-mouse secondary antibodies (Thermo Fisher Scientific) for 2 h at room temperature (RT). To stain the actin filaments of the cell, tissue was incubated in 0.2% (v/v) Phalloidin (Alexa Fluor 546 Phalloidin, Thermo Fisher Scientific) for 1 h at RT. Finally, the nuclei were stained by incubating the tissue in 0.1% (v/v) Hoechst 33342 (Thermo Fisher Scientific) for 15 min at RT.

Statistical Analysis: The data are presented as mean $\pm$ standard error of mean. The comparison of means was conducted using an F-test to confirm homoscedasticity, followed by an unpaired t-test assuming equal variance. For multiple comparisons, one-way ANOVA via Tukey's test was used. For the comparison of muscle fiber diameter, two-sided Mann-Whitney test was conducted. For all statistical analysis, $P < 0.05$ was considered as statistically significant. All analyses were performed with Prism 10.2.2 (GraphPad) software.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

biofabrication, microfabrication, skeletal muscle, tissue engineering, weight training

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